

AOS-CX 10.14.xxxx SNMP/MIB Guide

All AOS-CX Series Switches



**Hewlett Packard
Enterprise**

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)
- HPE Aruba Networking 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, S0X44A, S3L75A, S3L76A, S3L77A)
- HPE Aruba Networking 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A, S0E48A #0D1, S1T83A, S1T83A #0D1)
- HPE Aruba Networking 8100 Switch Series (R9W94A, R9W95A, R9W96A, R9W97A)
- HPE Aruba Networking 8320 Switch Series (JL479A, JL579A, JL581A)
- HPE Aruba Networking 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- HPE Aruba Networking 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A, JL700C, JL701C, JL702C, JL703C, JL706C, JL707C, JL708C, JL709C, JL710C, JL711C, JL704C, JL705C, JL719C, JL718C, JL717C, JL720C, JL722C, JL721C)
- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)
- HPE Aruba Networking 9300 Switch Series (R9A29A, R9A30A, R8Z96A, S0F81A, S0F82A, S0F83A, S0F84A, S0F85A, S0F86A, S0F87A, S0F88A, S0F95A, S0F96A)
- HPE Aruba Networking 10000 Switch Series (R8P13A, R8P14A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ▪ <code><example-text></code> ▪ <code><example-text></code> ▪ <code>example-text</code> ▪ <code>example-text</code>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ▪ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ▪ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ▪ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ▪ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

switch(CONTEXT-NAME)#

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where **<VLAN-ID>** is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

member/slot/port

On the Aruba 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the Aruba 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the Aruba 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the Aruba 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the Aruba 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

On the Aruba 8xxx, 9300, and 10000 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.



If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the Aruba 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:

- *member*: 1.
- *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing, and modifying information about managed devices on IP networks.



AOS-CX switches support RFC3411, RFC3412, RFC3413, RFC3414, RFC3415, RFC3416, RFC3417, and RFC3418 for SNMP.

SNMP write: PoE write capabilities

PoE enable

The PoE enable is requested through SNMP to enable the PoE interface. The `admin_disable` SNMP value is updated to enable the PoE interface.

PoE disable

The PoE disable is requested through SNMP to disable the PoE interface. The `admin_disable` SNMP value is updated to disable the PoE interface.

PoE cycle

The PoE cycle is a feature where you can request a PoE port reset with a timeout ranging from 1 to 60 seconds. The PoE cycle is requested through the SNMP server to disable and enable a PoE interface with an input timeout of 1 to 60 seconds. The PoE handles the PoE disable and enable events when the SNMP value is updated for `admin_disable` correspondingly. It is a one-time operation.

PoE priority

PoE priority handles the power priority to decide the number of ports to be powered up according to the set priority. This SNMP request sets the PoE interface PoE priority to one of these three values:

- `critical`
- `high`
- `low` (The default priority is `low`)

SNMP write: VLAN write capabilities

VLAN Add or Delete

The index for the `ieee8021QBridgeVlanStaticTable` and `dot1qVlanStaticTable` is VLAN. To create or delete a VLAN, either configure:

- ieee8021QBridgeVlanStaticRowStatus Mib object in ieee8021QBridgeVlanStaticTable
- or
- dot1QBridgeVlanStaticRowStatus Mib object in dot1qVlanStaticTable

Set the value to 4 to create a new VLAN, and set the value to 6 to delete an existing VLAN.

Add or Delete Tagged Port(s) to a VLAN

The index for the ieee8021QBridgeVlanStaticTable and dot1qVlanStaticTable is VLAN. To set or clear port(s) as tagged members for the VLAN, either configure:

- ieee8021QBridgeVlanStaticEgressPorts Mib object in ieee8021QBridgeVlanStaticTable
- or
- dot1QBridgeVlanStaticEgressPorts Mib object in dot1qVlanStaticTable

Changes to a bit in this object affect per-port and per-VLAN registrar control.

Add or Delete Untagged Port(s) to a VLAN

The index for the ieee8021QBridgeVlanStaticTable and dot1qVlanStaticTable is VLAN. To set or clear port(s) as untagged members for the VLAN, either configure:

- ieee8021QBridgeVlanStaticUntaggedPorts Mib object in ieee8021QBridgeVlanStaticTable
- or
- dot1QBridgeVlanStaticEgressPorts Mib object in dot1qVlanStaticTable

Changes to a bit in this object affect the per-port, per-VLAN Registrar control.

Enable or Disable MVRP to Port

The index for the ieee8021QBridgePortVlanTable and dot1qPortVlanTable is port. To enable or disable port level MVRP status, either configure:

- ieee8021QBridgeMvrpEnabledStatus Mib object in ieee8021QBridgePortVlanTable
- or
- dot1QBridgeMvrpEnabledStatus Mib object in dot1qPortVlanTable

The value true(1) indicates that MVRP is enabled on the device and false(2) indicates that MVRP is disabled for all ports on the device.

Add untagged VLAN to a Port

The index for the ieee8021QBridgePortVlanTable and dot1qPortVlanTable is port. To set untagged vlan for the port, either configure:

- ieee8021QBridgePvid Mib object in ieee8021QBridgePortVlanTable
- or
- dot1QBridgePvids Mib object in dot1qPortVlanTable

SNMP write: Configurations

Prerequisites

The switch must be configured for external access (such as management interface and IP addressing) and SNMP enabled (such as SNMPv2 and SNMPv3).

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip static 10.10.10.4/24
switch(config)# snmp-server vrf mgmt
switch(config)# no snmpv3 security-level auth-privacy
switch(config)# snmpv3 user test auth md5 auth-pass plaintext password priv aes
priv-pass plaintext password access-level rw
```

SNMP set examples

The following examples are executed from an external client communicating through SNMP to the switch. They describes both command syntax and OID interpretations:

■ copy running-config startup-config

```
snmpset -v3 -t100 -u test -l authPriv -a md5 -A password -x aes -X password
10.10.10.4 1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 3
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 2
```

OID	Description
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4	Create operation.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 3	Set source type to RunningConfig.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 2	Set destination type to StartupConfig.

■ copy startup-config running-config

```
snmpset -v3 -t100 -u test -l authPriv -a md5 -A password -x aes -X password
10.10.10.4 1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 2
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 3
```

OID	Description
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4	Create operation.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 2	Set source type to StartupConfig.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 3	Set destination type to RunningConfig.

■ copy REMOTE-URL running-config

```
snmpset -v3 -t100 -u test -l authPriv -a md5 -A password -x aes -X password
10.10.10.4 1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 1
```

```

1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 3
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.4.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.6.5 i 1
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.7.5 s "file"
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.9.5 s "10.10.10.1"
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.12.5 s "mgmt"
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.10.5 s "user"
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.11.5 s "password"
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.13.5 i 1

```

OID	Description
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4	Create operation.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 1	Set source type to external file.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 3	Set destination type to RunningConfig.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.4.5 i 4	Set protocol to SFTP.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.6.5 i 1	Set file format to CLI.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.7.5 s "file"	Set file name.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.9.5 s "10.10.10.1"	Set IP from server.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.12.5 s "mgmt"	Set VRF.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.10.5 s "user"	Set username to authenticate, if applicable.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.11.5 s "password"	Set password to authenticate, if applicable.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.13.5 i 1	Enable Notification on completion, if required.

■ copy running-config checkpoint ckpt1

```

snmpset -v3 -t100 -u test -l authPriv -a md5 -A password -x aes -X password
10.10.10.4 1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 3
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 4
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.5.5 s "ckpt1"

```

OID	Description
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.18.5 i 4	Create operation.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.2.5 i 3	Set source type to RunningConfig.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.3.5 i 4	Set destination type to Checkpoint.
1.3.6.1.4.1.47196.4.1.1.3.20.1.0.1.1.5.5 s "ckpt1"	Set checkpoint name to ckpt1.

SNMP MIB view

SNMP MIB view is a group of view subtrees in the MIB hierarchy. A view subtree is identified by the pairing of an Object Identifier (OID) subtree value with a bit string mask value. Each MIB view is defined by the view subtrees that is included or excluded from the MIB view. You can use the MIB views to control the OID range that SNMPv3 users or SNMP v1/v2 community can access.

Configuring SNMP MIB view

The following parameters are required to configure the SNMP MIB view:

- View name - Specifies the name of the SNMP MIB view. View names can support up to a maximum of 32 alphanumeric characters.
- Type - Whether to include or exclude the view subtree or group of subtrees from the SNMP MIB view.
- OID - An OID string for the subtree to include or exclude from the SNMP MIB view.

For example, the system subtree is specified by the OID string .1.3.6.1.2.1.1.

- Mask - It is an OID mask. The mask is 47 characters in length.
 - The format is xx:xx.... (:). Each OID mask is 16 octets in length.
 - An octet is two hexadecimal characters separated by : (colon). Only hexadecimal characters are accepted in this field.

For example, OID mask FF:A0 is 11111111:10100000.

SNMP MIB view limitations

An snmpwalk with a community or v3 user attached to context takes precedence over SNMP MIB view. If the context is attached to the user or community, the SNMP MIB view configured to the v3 user or community will not take effect.

Example:

```
snmp view admin system ff included
snmp-server community admin view admin
snmpv3 context new vrf default community admin
snmpv3 user nw_user context new
snmpv3 user nw_user view admin
```

Aruba proprietary MIB names without :: (double colon) separated root MIB names are not supported for SNMP MIB view.

- The following examples are supported for SNMP MIB view:

```
snmp-server view user_view .1.3.6.1.4.1.47196.4.1.1.3.11.6.1.1.4 FF included
snmp-server view admin_1 ARUBAWIRED-MODULE-MIB::arubaWiredModuleName FF included
```

- The following example is not supported for SNMP MIB view:

```
snmp-server view new_view arubaWiredModuleName included
```

SNMP traps

Event log traps

When SNMP is configured, interface daemons event log messages for link-up and link-down events will be sent as traps.

Event log trap OID: 1.3.6.1.4.1.47196.4.1.1.3.4.1.1

Parameter	OID	Description
sysUpTimeInstance	1.3.6.1.2.1.1.3.0	Contains the uptime for the system in centiseconds
snmpTrapOID	1.3.6.1.6.3.1.1.4.1	Contains the OID for the event log trap
eventIndex	1.3.6.1.2.1.16.9.1.1.1	Contains an index that uniquely identifies an event
eventDescription	1.3.6.1.2.1.16.9.1.1.2	Contains the event log message

Link-up and link-down traps

Standard IF-MIB link-up and link-down traps will be sent on link-state change when an interface is configured using `trap link-status` or when `user_config:link_state_snmp_trap` is set to true. The trap sends the current information for ifindex, admin status, operational status, and interface name.

Link up trap OID: 1.3.6.1.6.3.1.1.5.4

Link down trap OID: 1.3.6.1.6.3.1.1.5.3

Parameter	OID	Description
sysUpTimeInstance	1.3.6.1.2.1.1.3.0	Contains the uptime for the system in centiseconds
snmpTrapOID	1.3.6.1.6.3.1.1.4.1	Contains the OID for the link up or linkdown trap
ifIndex	1.3.6.1.2.1.2.2.1.1.X	Contains the ifindex for the interface
ifAdminStatus	1.3.6.1.2.1.2.2.1.7.X	Contains the admin status for the interface
ifOperStatus	1.3.6.1.2.1.2.2.1.8.X	Contains the operational status for the interface
ifDescr	1.3.6.1.2.1.2.2.1.2.X	Contains the name for the interface

Configuring SNMP

SNMP agent provides read-write access for specific OIDs. Refer [OIDs that support SNMP read-write](#) for the list of OIDs that supports read-write operations.



Best practices is to use a 5-second (-t5) timeout in a scaled environment, when the network traffic is high or when there is significant load on the switch.

Procedure

1. SNMP is not enabled on the switch by default, unless the user enables it over any available VRF or with the **default/mgmt** VRF using the command **snmp-server vrf <NAME>**. For example, use the

command **snmp-server vrf mgmt** to enable SNMP on the management interface. Use the command **snmp-server vrf default** to enable SNMP on the default VRF. Use the command **snmp-server vrf <USERDEFINED_VRF_NAME>** to enable SNMP on the user created VRF.

2. Set the system contact, location, and description for the switch with the following commands:
 - `snmp-server system-contact`
 - `snmp-server system-location`
 - `snmp-server system-description`

You can also set the system location and system contact values using SNMP.

3. If required, change the default SNMP port on which the agent listens for requests with the command **snmp-server agent-port**.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command **snmp-server community**.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command **snmp-server host**.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command **snmpv3 user <V3_USERNAME> context <CONTEXT_NAME>**.
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command **snmpv3 context <CONTEXT_NAME> vrf <VRF_NAME> community <COMMUNITY_NAME>**.
8. Review your SNMP configuration settings with the following commands:
 - `show snmp agent-port`
 - `show snmp community`
 - `show snmp system`
 - `show snmpv3 context`
 - `show snmp trap`
 - `show snmp vrf`
 - `show snmpv3 users`
 - `show tech snmp`

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF **mgmt**).
- Sets the contact, location, and description for the switch to: **JaniceM, Building2, LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user **Admin** with a context named **newContext**.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des  
priv-pass plaintext myprivpass  
switch(config)# snmpv3 user Admin context newContext
```

The CIPT MIB does not support the SNMP SET operation for the following objects:

- arubaWiredCiptEnable
- arubaWiredCiptProbeEnable
- arubaWiredCiptVlanConfig
- arubaWiredCiptPortEnable
- arubaWiredCiptPortUpdateInterval
- arubaWiredCiptPortClientLimit



SNMP commands

event-trap-enable

```
event-trap-enable  
no event-trap-enable
```

Description

Enables the notification of events to be sent as traps to the SNMP management stations. It is enabled by default.

The **no** form of this command disables the event traps.

Examples

Enabling the event traps:

```
switch(config)# event-trap-enable
```

Disabling the event traps:

```
switch(config)# no event-trap-enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp trap enable

```
lldp trap enable  
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.



LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
```

```
LLDP Global Configuration  
=====
```

```
LLDP Enabled           : No  
LLDP Transmit Interval : 30  
LLDP Hold Time Multiplier : 4  
LLDP Transmit Delay Interval : 2  
LLDP Reinit Timer Interval : 2  
LLDP Trap Enabled      : No
```

```
TLVs Advertised
```

```
=====
```

```
Management Address  
Port Description  
Port VLAN-ID  
System Description  
System Name
```

```
LLDP Port Configuration
```

```
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes

1/1/6	Yes	Yes	Yes
.....			
.....			
mgmt	Yes	Yes	Yes

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1

LLDP Global Configuration
=====
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No

LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
1/1/1         Yes             Yes             Yes
```

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt

LLDP Global Configuration
=====
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : Yes

LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config and config-if	Administrators or local user group members with execution rights for this command.

mac-notify traps

```
mac-notify traps {aged | learned | moved | removed}  
no mac-notify traps {aged | learned | moved | removed}
```

Description

Configures a Layer 2 or VXLAN interface to generate SNMP trap notifications for up to four different types of MAC address related events on the trunk or access in physical or lag interfaces.

MAC notification trap addition to or removal from an interface can be in any combination, quantity, or order. The addition of existing configured traps or removal of non-configured traps will be accepted and ignored.



The **mac-notify** feature must be enabled globally for any interface configurations to generate SNMP traps. Enabling **mac-notify traps** may impact the system performance on networks with a large number of mac-notify events.

The **no** form of this command removes the traps from the interface.

Parameter	Description
aged	Notifies when a MAC address aged out on the interface.
learned	Notifies when a MAC address is learned on the interface.
moved	Notifies when a MAC address moved from the interface.
removed	Notifies when a MAC address is removed from the interface.

MAC notification cannot be configured on a Layer 3 (routing) interface. A Layer 2 interface that is changed to a Layer 3 interface through the `routing` command will discard any existing MAC notification configurations.



When MACs are learned on VXLAN tunnels or **port-access port-security** enabled ports, the move scenario is handled by the EVPN/port-access feature respectively. It performs the move by deleting the MAC from the old port and installing it on the new port. In this scenario, MAC trap notifications, if enabled, will reflect that by producing **removed** and **learned** notifications.

Usage

- MAC notify trap will not generate for static MACs.
- **vsx-sync** is not supported. You must enable the MAC notify traps explicitly on secondary to ensure the traps are generated.
- For EVPN MAC move between the following interfaces, the respective event types are produced (not always removed or learned)
 - Port to port: **moved**
 - Port to tunnel: **removed/learned**
 - Tunnel to port: **removed/learned**
 - Tunnel to Tunnel: **moved**

Examples



MAC notification types and the associated events only apply to Layer 2 and VXLAN interfaces, hence routing might need to be disabled on the relevant interfaces.

Enable MAC notification traps within the SNMP module at a global level:

```
switch(config) # snmp-server trap

aaa-server-reachability-status  Enable SNMP trap for AAA server reachability
status
configuration-changes           Enable configuration changes traps
cpu-utilization                 Enable high CPU utilization traps
link-status                     Enable link status traps for all physical
interfaces
mac-notify                     Enable MAC table change notification traps
memory-utilization              Enable high memory utilization traps
module                         Enable module event traps
port-security                   Enable port-security violation traps.
                                (Default: enable)
rmon-events                     Enable RMON event traps
snmp                           Enable snmp traps
```

For more information, see [snmp-server trap mac-notify](#).

Enabling the traps on an L2 interface:

```
switch(config) # interface 1/1/1
switch(config-if) # mac-notify traps learned
1/1/1 is not an L2 interface or tunnel
switch(config-if) # no routing
switch(config-if) # mac-notify traps learned removed
switch(config-if) # mac-notify traps moved
switch(config-if) # mac-notify traps aged
```

```
switch(config) # interface vxlan 1
switch(config-vxlan-if) # mac-notify traps learned removed
```

```
switch(config) # interface lag101
switch(config-if) # mac-notify traps removed
```

Disabling the learned and removed traps from the interface 1/1/1:

```
switch(config) # interface 1/1/1
switch(config-if) # no mac-notify traps learned removed
```

```
switch(config) # interface vxlan 1
switch(config-vxlan-if) # no mac-notify traps learned removed
```

Enable sending SNMP notifications for MAC table changes:

```
switch(config-vxlan-if)# mac-notify traps
  aged          Notify when a MAC address aged out on the interface
  learned       Notify when a MAC address was learned on the interface
  moved         Notify when a MAC address moved from the interface
  removed       Notify when a MAC address was removed from the interface
switch(config-vxlan-if)# mac-notify traps learned aged removed moved
```

Command History

Release	Modification
10.13.1000	Support for SNMP MAC notify traps on VXLAN tunnels.
10.10	Support for port access features with mac-notify added.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command. Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

rmon alarm

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
    falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-
type <ABSOLUTE|DELTA>]
no rmon alarm [index <INDEX>]
```

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The **no** form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.
snmp-oid <SNMP-OID>	Specifies the SNMP MIB object to be monitored by RMON.
rising-threshold <RISING-THRESHOLD>	Specifies the upper threshold value for the RMON alarm.

Parameter	Description
falling-threshold <FALLING-THRESHOLD>	Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.
sample-interval <SAMPLE-INTERVAL>	Sample interval in seconds. Default: 30.
sample-type <ABSOLUTE DELTA>	Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Examples

Configuring RMON for the MIB object **ifOutErrors.15** with an index **1**, rising threshold of **2147483647** and falling threshold of **-2134** using **absolute** sampling for a sample interval of **100** seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold
2147483647
falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

rmon alarm {enable | disable} {index | all}

```
rmon alarm {enable | disable} {index <INDEX> | all}
no rmon alarm [enable | disable] [index <INDEX> | all]
```

Description

Enables and disables the RMON alarm and its index. RMON alarm is enabled by default.

Parameter	Description
enable	Enables the RMON alarm index
disable	Disables the RMON alarm index.
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.

Parameter	Description
all	Specifies all the RMON alarms.

Examples

Enabling or disabling all the RMON alarm:

```
switch(config)# rmon alarm enable all
switch(config)# rmon alarm disable all
```

Enabling or disabling RMON alarm by index:

```
switch(config)# rmon alarm enable index 1
switch(config)# rmon alarm disable index 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show configuration-changes trap

```
show configuration-changes trap
```

Description

Shows the SNMP configuration changes trap settings.

Example

Showing the SNMP configuration changes trap:

```
switch# show configuration-changes trap

SNMP Configuration changes trap : Enabled
^^^
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify

show mac-notify

Description

Displays whether the MAC notification feature in the SNMP module is enabled or not. It also displays the trap notification types configured on the Layer 2 ports in the system.

Examples

Showing the MAC notification configuration on all configured ports in the system:

```
switch# show mac-notify

MAC notification global setting : Enabled

Port          Enabled Traps
-----
1/1/1         aged learned moved
1/1/5         moved
lag101        removed
lag104        aged learned moved removed
...
...
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify port

show mac-notify [port <PORTS>]

Description

Displays the MAC notification configuration on a range of ports.

Parameter	Description
[port <PORTS>]	Specifies a port, range of ports, or list of ports.

Examples

Showing the MAC notification configuration on a range of ports:

```
switch(config)# show mac-notify port 1/1/1,1/1/3,1/1/5,lag101-lag104

MAC notification global Setting: Enabled

Port          Enabled Traps
-----
1/1/1         aged learned moved
1/1/3         --
1/1/5         moved
lag101        removed
lag102        --
lag103        --
lag104        aged learned moved removed
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show rmon alarm

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index          : 1
```

```

Enabled           : true
Status            : valid
MIB object        : ifOutErrors.15
Sample type       : delta
Sampling interval  : 6535 seconds
Rising threshold  : 100
Falling threshold : 10
Last sampled value : 0
Last sample time  : 2020-09-21 05:58:11

Index            : 3
Enabled          : true
Status           : invalid
MIB object        : IF-MIB::ifDescr.19
Sample type       : absolute
Sampling interval  : 10000 seconds
Rising threshold  : 4000
Falling threshold : 10
Last sampled value : 0

```

Showing RMON alarm with alarm index 1:

```

switch# show rmon alarm index 1
Index           : 1
Enabled         : true
Status          : valid
MIB object       : ifOutErrors.15
Sample type      : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11

```

Showing disabled RMON alarm information:

```

switch# show rmon alarm
Index           : 1
Enabled         : false
Status          : valid
MIB object       : ifOutErrors.15
Sample type      : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index           : 3
Enabled         : false
Status          : invalid
MIB object       : IF-MIB::ifDescr.19
Sample type      : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show snmp agent-port

show snmp agent-port

Description

Displays SNMP agent UDP port number.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp community

show snmp community

Description

Displays a list of all configured SNMPv1/v2c communities.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default **public** community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

```
switch#show snmp community
```

```
SNMP-COMMUNITIES
```

Community	Access-level	ACL Name	ACL Type	View
private	ro	my_acl	ipv4	view1
private	ro	my_acl	ipv6	none
private2	rw	new_Acl	ipv6	view2
private3	rw	none	none	none

When the switch is configured to use [SNMPv3 only](#), the output of the **show snmp community** command displays the message **SNMP v1/v2c is disabled while snmpv3-only mode is configured**:

```
switch# show snmp community
```

Community	Access-level	ACL Name	ACL Type
View			

```
SNMP v1/v2c is disabled while snmpv3-only mode is configured
```

Command History

Release	Modification
10.14	The output of this command now displays an error message when the switch is in SNMPv3-only mode.
10.10	Output has been updated with SNMP view details. A <i>View</i> column is added to the command output.
10.08	Added <i>ACL Type</i> column to the command output.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show snmp system

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
-----
System description : Aggregation router
System location : Main lab
System contact : John Smith, Lab Admin
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp trap

show snmp trap

Description

Displays all configured SNMP traps/informs receivers.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
```

HOST	PORT	TYPE	VER	COMMUNITY/USER	NAME	VRF	NOTIFICATION TYPES
-							
10.10.10.10	162	trap	v1	public		default	bgp
10.10.10.10	162	inform	v2c	public		default	bgp, ospf, fan,
mstp							
10.10.10.10	162	inform	v3	name		default	

Command History

Release	Modification
10.14	Updated the example output.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp views

show snmp views

Description

Displays the list of all the configured SNMP views.

Usage

The following table contains the status and its description of the configured SNMP views:

Status	Description
pending_validation	Default value that indicates SNMP view is yet to be validated.
operational	OID and mask validated.
invalid	Invalid OID/mask.
failed	Validation failed for reasons other than OID/mask.

Examples

Displaying the list of all the configured SNMP views:

```
switch# show snmp views
-----
SNMP MIB Views
-----
View      : new
OID Tree: sysUpTime.0
Mask      : ff
Type      : included
Status    : pending_validation

View      : admin
OID Tree: ifIndex.1
Mask      : ff:a0
Type      : included
Status    : operational

View      : user
OID Tree: sysb
Mask      : none
Type      : excluded
Status    : invalid

View      : admin
OID Tree: .1.3.6.1.2.1.1
Mask      : none
```

```
Type    : excluded
Status  : operational
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp vrf

```
show snmp vrf
```

Description

Displays the VRF on which the SNMP agent service is running.

Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
-----
mgmt
default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 context

```
show snmpv3 context
```

Description

Displays all configured SNMP contexts.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
-----
name                                vrf                                community
-----
contextA                           default                            private
contextB                           vrf_A                             public
```

```
switch# show snmpv3 context
-----
Name                                vrf                                Community                        type[Instance_id]
-----
A                                   default                            public                           vrf
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 engine-id

```
show snmpv3 engine-id
```

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Example

Displaying the configured SNMPv3 engine-id:

```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 security-level

```
show snmpv3 security-level
```

Description

Displays the configured SNMPv3 security level.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 users

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
```

User	AuthMode	PrivMode	Status	Context	Access-level	View
name	md5	none	Enabled	context2 context1 context3	ro	view1
name2	none	none	Disabled	none	ro	view2
name3	none	none	Disabled	none	ro	none

Command History

Release	Modification
10.10	Output has been updated with SNMP view details. A <i>View</i> column is added to the command output.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

snmp-server agent-port

```
snmp-server agent-port <PORT>  
no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The **no** form of this command sets the SNMP master agent port to the default value.

Parameter	Description
<PORT>	Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Examples

Setting the SNMP master agent port to **2000**:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server community

```
snmp-server community <STRING>  
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is like a password that controls read/write access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The **no** form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value **public** is automatically defined.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Subcommands

```
access-level {ro | rw}  
no access-level {ro | rw}
```

This subcommand changes the access level of the SNMP community. The default access level is read-only (**ro**).

The **no** form of this subcommand changes the access level of the community to default.

Parameter	Description
ro	Specifies Read-Only access with the SNMP community.
rw	Specifies Read-Write access with the SNMP community.

```
access-list {ip | ipv6} <ACL-NAME>  
no access-list {ip | ipv6} <ACL-NAME>
```

This subcommand associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The **no** form of this subcommand removes association of the ACL with the SNMP community.

Parameter	Description
ip	Specifies the IPv4 ACL type.
ipv6	Specifies the IPv6 ACL type.
<ACL-NAME>	Specifies the ACL name. It supports a maximum of 64 characters.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config) # snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config) # no snmp-server community private
```

Configuring the access level for the SMNP community to read-only:

```
switch(config-community) # access-level ro
```

Changing the access level of the SNMP community to default:

```
switch(config-community) # no access-level rw
```

Associating an IPv4 ACL named **my_acl** with the SMNP community:

```
switch(config-community) # access-list ip my_acl
```

Removing the associated IPv4 ACL named **my_acl** from the SNMP community:

```
switch(config-community) # no access-list ip my_acl
```

Configuration supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 permit any 4.4.4.4 4.4.4.1
  20 permit any 3.3.3.3 3.3.3.1
access-list ipv6 ipv6_acl
  10 permit any 2001::2 2001::1
  20 permit any 3001::2 3001::1
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```

Configuration not supported for SNMP ACL:

```
access-list ip ipv4_acl
 10 deny any 6.6.6.6 6.6.6.1
access-list ipv6 ipv6_acl
 10 deny any 6001::6 6000::1
snmp-server vrf default
snmp-server community my_comm_1
access-list ip ipv4_acl
access-list ipv6 ipv6_acl
```



hitcounts for SNMP ACL will not be incremented.

Example: `show access-list hitcounts ip all` will not show the hit count of SNMP ACL.

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config config-community	Administrators or local user group members with execution rights for this command.

snmp-server community view

```
snmp-server community <STRING> [view <VIEW-NAME>]
no snmp-server community <STRING> [view <VIEW-NAME>]
```

Description

Associates an SNMP MIB view with the SNMP community.

The **no** form of this command removes the associated SNMP MIB view from the SNMP community.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.
<VIEW-NAME>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Configuring the SNMPv1/SNMPv2c community:

```
switch(config)# snmp-server community my_community
switch(config-community)#
```


Adding SNMP MIB view to the SNMP community:

```
switch(config-community) # view name1
```

Removing SNMP MIB view from the SNMP community:

```
switch(config-community) # no view name1
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config config-community	Administrators or local user group members with execution rights for this command.

snmp-server historical-counters-monitor

```
snmp-server historical-counters-monitor  
no snmp-server historical-counters-monitor
```

Description

Enables the Remote Network Monitoring agent (**rmond**) to start collecting historical interface statistics. The **no** form of this command stops the historical interface statistics collection.

Example

Enabling the `rmond` agent to start historical interface statistics collection:

```
switch(config) # snmp-server historical-counters-monitor
```

Disabling the `rmond` agent to stop historical interface statistics collection:

```
switch(config) # no snmp-server historical-counters-monitor
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

snmp-server host

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user
<NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user
<NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<IPv6-ADDR>	Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).
trap version <VERSION>	Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1 , v2c or v3 .
inform version v2c	Specifies the inform notification type for SNMPv2c.
trap version v3	Specifies the trap notification type for SNMPv3.
user <NAME>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
community <STRING>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public .
<UDP-PORT>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

Parameter	Description
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests.
<code><notification-type></code>	Specifies the type of notification to be sent to the trap receiver. If no type is specified, all notifications are sent. The supported notification types are: ▪ aaa-server ▪ alarm ▪ bgp ▪ card ▪ config ▪ entity ▪ fan ▪ interface ▪ lldp ▪ loop-protect ▪ mac-notify ▪ mstp ▪ mvrp ▪ ospf ▪ ospfv3 ▪ port-security ▪ power ▪ power-ethernet ▪ rmon ▪ rpvst ▪ stp ▪ temperature ▪ vrrp ▪ vsf ▪ vsx

Examples

```

switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host a:b::c:d trap version v2c community public port
5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community public

```

```

port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# snmp-server host a:b::c:d inform version v2c community public
switch(config)# no snmp-server host a:b::c:d inform version v2c community public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default
switch(config)# snmp-server host a:b::c:d inform version v2c community public port
5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port
2000
switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port 2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin port 2000

```

SNMP trap notification type examples:

```

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
notification-type bgp fan interface power entity
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
notification-type bgp
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp fan interface power-ethernet
switch(config)# no snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp interface
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type ?
  aaa-server      Sends AAA notifications.
  alarm           Sends Alarm notifications.
  bgp             Sends Border Gateway Protocol (BGP) state change notifications.
  card            Sends Card notifications.
  config          Sends Configuration change notifications.
  entity          Sends Entity notifications.
  fan             Sends Fan notifications.
  interface       Sends Interface notifications.
  lldp            Sends Link Layer Discovery Protocol (LLDP) notifications.
  loop-protect    Sends Loop Protect notifications.
  mac-notify      Sends MAC Notify notifications.
  mstp            Sends Multiple Spanning Tree Protocol (MSTP) notifications.
  mvrp            Sends Multiple VLAN Registration Protocol (MVRP) notifications.
  ospf            Sends Open Shortest Path First (OSPFv2) notifications.
  ospfv3          Sends Open Shortest Path First version 3 (OSPFv3) notifications.
  port-security   Sends Port Security notifications.
  power           Sends Power notifications.
  power-ethernet  Sends Power over Ethernet (PoE) notifications.
  rmon            Sends Remote Network Monitoring (RMON) notifications.
  rpvst           Sends Rapid Per VLAN Spanning Tree (RPVST) notifications.

```

snmp notifications.	Sends Sends Simple Network Management Protocol (SNMP)
stp	Sends Spanning Tree Protocol (STP) notifications.
temperature	Sends Temperature notifications.
vrrp	Sends Virtual Router Redundancy Protocol (VRRP) notifications.
vsf	Sends Virtual Switching Framework (VSF) notifications.
vsx	Sends Virtual System Extension (VSX) notifications.

Show SNMP trap example:

```
switch(config)# show snmp trap

-----
-----

Host          Port  Type  Version Community-Name/User-Name      vrf
Notification Types

-----
-----

192.168.12.239 162   trap  v3      kgbhat1                        mgmt   fan
192.168.16.190 162   trap  v2c     public                        mgmt
```

config, entity, interface, rmon

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server response-source

```
snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]
no snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]
```

Description

Configures the source interface or IP address for sending SNMP responses. Each SNMP can independently have its own unique response source IP address.

The **no** form of this command removes the source interface name or IP address for sending SNMP responses.



- It is recommended to use the loopback interface or ip address of the loopback interface as the response source. If a device does not support a loopback interface, then configure SVI interface or SVI IP address as the response source.
- The active gateway IP address cannot be configured as the response source.
- It is recommended to limit the maximum number of response source to five.
- The interface used for the response source should be in the up state. If the interface is down, the default source IP will be used.
- The use of **udp6** is mandatory for IPv6 SNMP operations. For example, you can use the following syntax:
snmpwalk -v2c -c public -m ALL udp6:[2100::2] .1.3.6.1.2.1.1.

Parameter	Description
<code>interface <IF-NAME></code>	Specifies the source interface name. The interface can be a physical interface, loopback interface, or VLAN interface.
<code><IPv4-ADDRESS></code>	Specifies the IPv4 address of the source interface for the SNMP response.
<code><IPv6-ADDRESS></code>	Specifies the IPv6 address of the source interface for the SNMP response.
<code>vrf <VRF-NAME></code>	Specifies the VRF name associated to the source interface for the SNMP response.

Examples

Configuring a response source for the interface **1/1/12**:

```
switch(config)# snmp-server response-source interface 1/1/12 vrf red
```

Configuring a response source for interface **loopback10**:

```
switch(config)# snmp-server response-source interface loopback10 vrf red
```

Configuring a response source for the IPv4 address **10.0.0.1**:

```
switch(config)# snmp-server response-source 10.0.0.1 vrf sample
```

Configuring a response source for the IPv6 address **2001::1**:

```
switch(config)# snmp-server response-source 2001::1 vrf default
```

Command History

Release	Modification
10.13	Added support for IPv6 address.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server snmpv3-only

```
snmp-server snmpv3-only
no snmp-server snmpv3-only
```

Description

Accepts SNMPv3 messages only, SNMPv1 and SNMPv2c will be disabled. By default SNMPv1, SNMPv2c and SNMPv3 will all be enabled.

The **no** form of this command restores the default setting and reenables SNMPv1 and SNMPv2c .

Examples

Configuring SNMPv3 messages only, and disabling SNMPv1 and SNMPv2c:

```
switch(config)# snmp-server snmpv3-only
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100 6200 6300 6400 8100 8320 8325 8360 8400 9300 10000	config	Administrators or local user group members with execution rights for this command.

snmp-server system-contact

```
snmp-server system-contact <INFO>
no snmp-server system-contact [<INFO>]
```

Description

Sets SNMP contact information.

The **no** form of this command removes the SNMP contact information.

Parameter	Description
<INFO>	Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-description

```
snmp-server system-description <DESCRIPTION>
no snmp-server system-description
```

Description

Sets the SNMP system description.

The **no** form of this command removes the SNMP system description.

Parameter	Description
<DESCRIPTION>	Specifies the SNMP system description. Typical content to include would be the full name and version of the following:

Parameter	Description
	- Hardware type of the system - Software operating system - Networking software Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-location

```
snmp-server system-location <INFO>
no snmp-server system-location
```

Description

Sets the SNMP location information.

The **no** form of this command removes the SNMP location information.

Parameter	Description
<INFO>	Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap

```
snmp-server trap {cpu-utilization | memory-utilization | rmon-events}  
no snmp-server trap {cpu-utilization | memory-utilization | rmon-events}
```

Description

Enables the SNMP traps. The SNMP traps are enabled by default.

The **no** form of this command disables the SNMP traps.

Parameter	Description
cpu-utilization	Enables the CPU utilization traps.
memory-utilization	Enables the memory utilization traps.
rmon-events	Enables the RMON event traps.

Examples

Enabling the SNMP traps:

```
switch(config)# snmp-server trap cpu-utilization  
switch(config)# snmp-server trap memory-utilization  
switch(config)# snmp-server trap rmon-events
```

Disabling the SNMP traps:

```
switch(config)# no snmp-server trap cpu-utilization  
switch(config)# no snmp-server trap memory-utilization  
switch(config)# no snmp-server trap rmon-events
```

Displaying the SNMP trap configuration:

```
switch(config)# show running-config all | inc snmp  
snmp-server trap rmon-events  
snmp-server trap cpu-utilization  
snmp-server trap memory-utilization
```

Displaying CPU and Memory usage:

```
switch(config)# show system  
Hostname           : XXXX  
System Description : XX.10.07.0001CI  
System Contact      :  
System Location     :  
Vendor              : Aruba  
Product Name        : JLXXXX XXXX Base Chassis/3xFT/18xFans/Cbl Mgr/X462 Bundle  
Chassis Serial Nbr  : SG6Z009068  
Base MAC Address    : f40343-806400  
AOS-CX Version      : XX.10.07.0001CI  
Time Zone           : UTC  
Up Time             : 8 minutes  
CPU Util (%)        : 1  
Memory Usage (%)    : 10
```



cpu-utilization and **memory-utilization** are not supported in 4100i, 6000 and 6100 switch series. For more information on features that use this command, refer to the SNMP/MIB Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100 6200 6300 6400 8100 8320 8325 8360 8400 9300 10000	config	Administrators or local user group members with execution rights for this command.

snmp-server trap aaa-server-reachability-status

```
snmp-server trap aaa-server-reachability-status  
no snmp-server trap aaa-server-reachability-status
```

Description

Enables the SNMP trap for AAA server status. When enabled, traps are sent whenever AAA server (RADIUS, TACACS) status changes from reachable to unreachable and vice versa.

The **no** form of this command disables sending SNMP trap for AAA server status.

Examples

Enabling the SNMP trap for AAA server status:

```
switch(config)# snmp-server trap aaa-server-reachability-status
```

Disabling the SNMP trap for AAA server status:

```
switch(config)# no snmp-server trap aaa-server-reachability-status
```

Command History

Release	Modification
10.10	Command introduced on 4100i, 6000, 6100, 8100, 8320, 8325, 8360, 8400, 9300, and 10000 switches.
10.09	Command introduced on 6200, 6300 and 6400 switches.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap configuration-changes

```
snmp-server trap configuration-changes  
no snmp-server trap configuration-changes
```

Description

Enables sending SNMP traps whenever the configuration changes. Configuration trap generation is disabled by default.

The **no** form of this command disables sending SNMP traps for configuration changes.

Parameter	Description
configuration-changes	Specifies SNMP traps for configuration changes.

Examples

Enabling the SNMP traps for configuration changes:

```
switch(config)# snmp-server trap configuration-changes
```

Disabling the SNMP traps for configuration changes:

```
switch(config)# no snmp-server trap configuration-changes
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap mac-notify

```
snmp-server trap mac-notify  
no snmp-server trap mac-notify
```

Description

Enables the MAC notification traps within the SNMP module at a global level. When enabled, traps are sent for interfaces that are configured for MAC notification events.

The **no** form of this command disables sending MAC notification traps at a global level. When disabled, existing **mac-notify** interface configuration is preserved but MAC notification events on configured interfaces will not cause SNMP traps to be transmitted.

Examples

Enabling the SNMP MAC notification feature in the system globally:

```
switch(config)# snmp-server trap mac-notify
```

Disabling the SNMP MAC notification feature in the system globally:

```
switch(config)# no snmp-server trap mac-notify
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap module

```
snmp-server trap module
no snmp-server trap module
```

Description

Enables SNMP trap generation for modules. Module trap generation is enabled by default. Generates the module event traps whenever a modular line or fabric card changes state, which includes inserted, removed, ready, and down, as well as when a modular card is unrecognized.

The **no** form of this command disables the SNMP trap generation for module events.

Parameter	Description
module	Specifies SNMP traps for module events.

Examples

Enabling the SNMP traps for modules:

```
switch(config)# snmp-server trap module
```

Disabling the SNMP traps for modules:

```
switch(config)# no snmp-server trap module
```

```
switch(config)# show running-config
no snmp-server trap module
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6400 8400	config	Administrators or local user group members with execution rights for this command.

snmp-server trap port-security

```
snmp-server trap port-security
no snmp-server trap port-security
```

Description

Enables SNMP port-security violation traps on the system. Port-security violation traps are enabled by default.

The **no** form of this command disables the SNMP port-security violation traps on the system.

Parameter	Description
port-security	Specifies SNMP traps for port-security.

Examples

Enabling the SNMP port-security violation traps on the system:

```
switch(config)# snmp-server trap port-security
```

Disabling the SNMP port-security violation traps on the system:

```
switch(config)# no snmp-server trap port-security
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100 6200 6300 6400 8100 8360	config	Administrators or local user group members with execution rights for this command.

snmp-server trap snmp

```
snmp-server trap snmp {authentication | coldstart | warmstart} [vrf <VRF_NAME>]
no snmp-server trap snmp {authentication | coldstart | warmstart} [vrf <VRF_NAME>]
```

Description

Enables SNMPv2 MIB traps. The SNMPv2 traps are disabled by default.

The **no** form of this command disables the SNMPv2 MIB traps.

SNMPv2 MIB supports the following traps:

- **authentication**: Authentication trap is sent when the SNMP server receives a protocol message that is not properly authenticated.
- **coldstart**: A coldstart trap is sent when the switch reboots.
- **warmstart**: A warmstart trap is sent when there is a user intervention to enable or disable the SNMP service on the switch.



SNMPv2 Authentication traps do not support source IP configuration.

Parameter	Description
authentication	Enables the authentication traps.
coldstart	Enables the coldstart traps.
warmstart	Enables the warmstart traps.
<VRF_NAME>	Specifies the VRF name. Enables the SNMPv2 traps for a VRF.

Examples

Enabling all SNMPv2 traps:

```
switch(config)# snmp-server trap snmp
```

Enabling only SNMPv2 authentication traps:

```
switch(config)# snmp-server trap snmp authentication
```

Disabling all SNMP traps:

```
switch(config)# no snmp-server trap snmp
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap-source interface vrf

```
snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
no snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

Description

Configures SNMP trap source interface or IP address for a VRF.

The **no** form of this command removes the SNMP **trap-source** configuration for a VRF.

Parameter	Description
<IF-NAME>	Specifies the source interface name. Interface name can be physical interface, loopback interface, LAG interface, or VLAN interface.
<IPv4-Address>	Specifies the IPv4 address of source interface for the SNMP trap.
<IPv6-Address>	Specifies the IPv6 address of source interface for the SNMP trap.
<VRF-NAME>	Specifies the name of a VRF associated to the source interface for the SNMP trap.

Examples

Configuring SNMP trap source interface for a VRF.

```
switch(config)# snmp-server trap-source interface 1/1/12 vrf sample
switch(config)# snmp-server trap-source interface loopback10 vrf sample
switch(config)# snmp-server trap-source interface vlan23 vrf sample
```

Configuring SNMP trap source IP address for a VRF.

```
switch(config)# snmp-server trap-source 10.0.0.1 vrf red
switch(config)# snmp-server trap-source 1001::1 vrf red
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap vsx

```
snmp-server trap vsx
no snmp-server trap vsx
```

Description

Enables sending the SNMP traps for VSX related events. VSX trap generation is disabled by default. The **no** form of this command disables sending the SNMP traps for VSX related events. The trap support is available for the following VSX events:

- ISL up and down
- KA up and down
- MCLAG up and down

Parameter	Description
vsx	Specifies SNMP traps for VSX events.

Examples

Enabling the VSX traps:

```
switch(config)# snmp-server trap vsx
```

```
switch(config)# show vsx configuration trap
SNMP traps : Enabled
```

Disabling the VSX traps:

```
switch(config)# no snmp-server trap vsx
```

```
switch(config)# show vsx configuration trap
SNMP traps : Disabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server view

```
snmp-server view <VIEWNAME> <OID_TREE> [<MASK>] <included/excluded>
no snmp-server view <VIEWNAME> <OID_TREE> [<MASK>] <included/excluded>
```

Description

Configures an SNMP MIB view.

The **no** form of this command removes the specified SNMP MIB view.

Parameter	Description
<VIEWNAME>	Specifies the name of the SNMP MIB view. Supports up to a maximum of 32 characters.
<OID_TREE>	Specifies the OID tree to be included or excluded in SNMP MIB

Parameter	Description
	view.
<MASK>	Specifies the OID mask value. The values must be in hexadecimal character separated with : (colon).
<included/excluded>	Specifies the OID tree that is included in or excluded from the SNMP MIB view.

Usage

You can configure a maximum of 50 SNMP MIB views. The following VTY message is displayed when the configuration exceeds the maximum SNMP MIB views:

```
switch(config)# snmp-server view name51 1.3.6.1.2.1.1 fe:00 included
Configuration failed: Maximum allowed views are configured.
```

Examples

Configuring the SNMP MIB views:

```
switch(config)# snmp-server view name1 .1.3.6.1.2.1.2.2.1.1.1 FF:A0 included
switch(config)# snmp-server view name2 IF-MIB::ifindex included
switch(config)# snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Removing an SNMP MIB view:

```
switch(config)# no snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 4100i, 6000, and 6100 only support default VRF. **The SNMP agent can listen on multiple VRFs.**

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Examples

Configuring the SNMP agent to listen on VRF **default**.

```
switch(config)# snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF **mgmt**.

```
switch(config)# snmp-server vrf mgmt
```

Configuring the SNMP agent to listen on used-defined VRF **myvrf**.

```
switch(config)# snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF **default**.

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 context

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]  
no snmpv3 context <NAME> [vrf <VRF-NAME>] [community <STRING>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

Parameter	Description
<NAME>	Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).
vrf <VRF-NAME>	Specifies the VRF associated with the context. Default: default.
community <STRING>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 engine-id

```
snmpv3 engine-id <ENGINE-ID>
no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The **no** form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Parameter	Description
<code><ENGINE-ID></code>	SNMPv3 SNMP engine-id in colon separated hexadecimal notation.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
WORD SNMPv3 snmp engine-id in colon seperated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmpv3 security-level

```
snmpv3 security-level {auth | auth-privacy}
no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SMNPv3 users defined by the command `snmpv3 user` are able to connect.

The **no** form of this command changes the security level as follows:

- **no snmpv3 security-level auth:** Sets the security level to **auth-privacy**.
- **no snmpv3 security-level auth-privacy:** Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Parameter	Description
<code>auth</code>	SNMPv3 users that support authentication, or authentication and privacy are allowed.
<code>auth-privacy</code>	Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

```
snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]

no snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command **snmpv3 security-level**) determines which users are allowed to authenticate.

The **no** form of this command removes the specified SNMPv3 user.



When updating the authentication protocols and privacy protocols for the existing SNMPv3 users, you must also update the access level. Otherwise, the access level will be set to read-only.

Parameter	Description
<code><NAME></code>	Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).
<code>access-level</code>	Configures the access level for the SNMPv3 user: ▪ ro: Allow read-only access for the SNMPv3 user ▪ rw: Allow read-write access for the SNMPv3 user
<code>auth <AUTH-PROTO></code>	Sets the authentication protocol used to validate user logins. Supported protocols are md5 , sha , sha224 , sha256 , sha384 , and sha512 .
<code>auth-pass [{plaintext ciphertext} <AUTH-PASS>]</code>	Specifies the SNMPv3 user authentication password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches. NOTE: Authentication passwords that include special characters must be enclosed in single quotation marks (''). For example, 'auth-pwd20246!@#'.
<code>priv <PRIV-PROTO></code>	Sets the SNMPv3 privacy protocol (encryption method). Supported privacy protocols are aes , aes192 , aes256 , and des .
<code>priv-pass [{plaintext ciphertext} <PRIV-PASS>]</code>	Specifies the SNMPv3 user privacy encryption password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches. NOTE: Authentication passwords that include special characters must be enclosed in single quotation marks (''). For example, 'priv-pwd20246!@#'.



When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.



When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
priv aes priv-pass
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
Enter the authentication password: *****
Re-Enter the authentication password: *****

Configure the privacy protocol (y/n)? y
Enter the privacy protocol (aes/des)? aes

Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the **show running-config** command to display switch configuration. Save a copy of the full **snmpv3 user** command (shown by **show running-config**). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX xx.xx.xx.xxxxxx
```

```

!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
    no shutdown
    ip dhcp
    vlan 1

```

On switch 2, execute the **snmpv3 user** command that you saved from switch 1 (as shown by **show running-config**). This creates the user on switch 2 with the same configuration.

```

switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext
AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=

```

The following command sets a read-write access level for an SNMPv3 user with the user name **user1**.

```

switch(config)# snmpv3 user user1 auth md5 auth-pass plaintext abc1234 access-
level rw

```

Command History

Release	Modification
10.13	Following authentication protocols are supported: sha224 , sha256 , sha384 , and sha512 . Following privacy protocols are supported: aes192 and aes256 .
10.09	The access-level parameter was introduced.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user view

```

snmpv3 user <USER-NAME> view <VIEW-NAME>
no snmpv3 user <USER-NAME> view <VIEW-NAME>

```

Description

Associates a user with an existing SNMP MIB view.

The **no** form of this command removes the associated user from the specified SNMP MIB view.

Parameter	Description
<code><USER-NAME></code>	Specifies the user name for the SNMP MIB view. Accepts a maximum of 32 characters.
<code><VIEWNAME></code>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Adding a user in the existing SNMP MIB view:

```
switch(config)# snmpv3 user nw-admin view my-nw-view
```

Removing the user from the SNMP MIB view:

```
switch(config)# no snmpv3 user nw-admin view my-nw-view
```

Attaching unconfigured or unknown SNMP view to an SNMPv3 user:

```
switch(config)# snmpv3 user nw-admin view myView
View myView is not configured.
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Entity MIB support

The Entity MIB, rfc 6933, allows network managers to retrieve physical containment and logical relationships for devices in the network. The entconfigChange trap is sent to configured SNMP-server hosts when a change occurs. The trap is configured to send notifications no more than once every 5 seconds. We will be supporting the Entity MIB for read-only.

Physical components that are supported include:

- Stack
- Chassis
- Fabric cards
- Fan trays
- Fans

- Line cards and their interfaces
- Management modules and the intake temperature sensor
- Power supplies

The slots for any removable component are also represented. The logical table of the Entity MIB represents configured VLANs and the associated ports. The entConfigChange trap/notification is sent to configured snmp-server hosts.

Location of the MIB files on the web

The MIB files for Aruba switches can be found on the [Aruba Service Portal](#). You can apply the various filters to filter by product series, software versions, and software release types.

Updated MIBs and Traps for AOS-CX 10.14

The following list contains the newly introduced, deprecated, and obsolete MIBs and Traps for each software feature. Software is provided along with the MIB and Traps supported name.

New MIB Objects Introduced in 10.14

Updated MIB file: ARUBAWIRED-INTERFACE-MIB.mib

The **ARUBAWIRED-INTERFACE-MIB.mib** MIB now includes a table with information about the rate of the physical ports of this device.

This table, **arubaWiredIfRateTable**, includes the following objects:

- **arubaWiredIfRateIndex**: An index that uniquely identifies a row in the physical port stat table.
- **arubaWiredIfRateInterval**: Rate collection interval in seconds.
- **arubaWiredIfRateInBytes**: Received bytes per second over the rate interval.
- **arubaWiredIfRateOutBytes**: Transmitted bytes per second over the rate interval.
- **arubaWiredIfRateInBcastPkts**: Received broadcast packets per second over the rate interval.
- **arubaWiredIfRateInMcastPkts**: Received multicast packets per second over the rate interval.
- **arubaWiredIfRateInUcastPkts**: Received unicast packets per second over the rate interval.
- **arubaWiredIfRateInTotalPkts**: Received packets per second over the rate interval.
- **arubaWiredIfRateOutBcastPkts**: Transmitted broadcast packets per second over the rate interval.
- **arubaWiredIfRateOutMcastPkts**: Transmitted multicast packets per second over the rate interval.
- **arubaWiredIfRateOutUcastPkts**: Transmitted unicast packets per second over the rate interval.
- **arubaWiredIfRateOutTotalPkts**: Transmitted packets per second over the rate interval.
- **arubaWiredIfRateInUtilization**: Inbound interface utilization over the rate interval.
- **arubaWiredIfRateOutUtilization**: Outbound interface utilization over the rate interval.

Updated MIBs for AOS-CX 10.14.1000

This MIB provides the detailed information of the transceivers modules plugged into the switch ports.

Updated MIB file: ARUBAWIRED-PM-MIB.mib

arubaWiredPmXcvrTable	
arubaWiredPmXcvrPortIfIndex	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.1	
arubaWiredPmXcvrPortDesc	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.2	
arubaWiredPmXcvrDescription	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.3	
arubaWiredPmXcvrProductNum	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.4	
arubaWiredPmXcvrSerialNum	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.5	
arubaWiredPmXcvrPartNum	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.6	
arubaWiredPmXcvrType	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.7	
arubaWiredPmXcvrConnectorType	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.8	
arubaWiredPmXcvrConnectorStatus	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.9	
arubaWiredPmXcvrConnectorStatusReason	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.10	
arubaWiredPmXcvrCableType	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.11	
arubaWiredPmXcvrWavelength	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.12	
arubaWiredPmXcvrSmfTxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.13	
arubaWiredPmXcvrMmfOm1TxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.14	
arubaWiredPmXcvrMmfOm2TxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.15	
arubaWiredPmXcvrMmfOm3TxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.16	
arubaWiredPmXcvrMmfOm4TxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.17	
arubaWiredPmXcvrMmfOm5TxDist	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.18	
arubaWiredPmXcvrCableLength	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.19	
arubaWiredPmXcvrMaxSpeed	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.20	
arubaWiredPmXcvrMaxPower	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.21	
arubaWiredPmXcvrAdapterType	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.22	
arubaWiredPmXcvrAdapterStatus	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.23	
arubaWiredPmXcvrAdapterStatusReason	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.24	
arubaWiredPmXcvrAdapterProductNum	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.25	
arubaWiredPmXcvrAdapterSerialNum	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.26	
arubaWiredPmXcvrThermalClass	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.27	
arubaWiredPmXcvrDiagnostics	->
1.3.6.1.4.1.47196.4.1.1.3.27.1.1.1.28	

OIDs that support SNMP read-write

The following table contains the OIDs that support SNMP read-write:

Software Feature	MIB File	OID
SNMPv2 System	SNMPv2-MIB.mib	- sysContact - sysName - sysLocation
PoE	ARUBAWIRED-POE.mib	arubaWiredPoePethPsePortPoECycle
	POWER-ETHERNET-MIB	pethPsePortAdminEnable
VLAN	IEEE8021QBridge-Mib	- ieee8021QBridgeVlanStaticEgressPorts - ieee8021QBridgeVlanStaticRowStatus - ieee8021QBridgeVlanStaticUntaggedPorts - ieee8021QBridgePvid - ieee8021QBridgeMvrpEnabledStatus
	Dot1QBridge-Mib	- Dot1QBridgeVlanStaticEgressPorts - Dot1qVlanStaticRowStatus - Dot1QBridgeVlanStaticUntaggedPorts - Dot1QBridgePvid - Dot1QBridgeMvrpEnabledStatus
Interface	IF-MIB	- ifAdminStatus - ifAlias
	ARUBAWIRED-INTERFACE-MIB	- arubaWiredInterfaceAutoneg - arubaWiredInterfaceDuplex - arubaWiredInterfaceSpeeds
Port Security	ARUBAWIRED-PORTSECURITY-MIB	- arubaWiredPortSecurityGlobalEnab - arubaWiredPortSecurityEnable - arubaWiredClientLimit - arubaWiredViolationAction - arubaWiredRecoveryTimer - arubaWiredShutdownRecovery - arubaWiredStickyEnable

OIDs that support SNMP read-create

The following table contains the OIDs that support SNMP read-create:

Software Feature	MIB File	OID
Port Security	ARUBAWIRED-PORTSECURITY-MIB	- arubaWiredClientMacVidList - arubaWiredMacAddrRowStatus

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.arubanetworks.com/support-services/
AOS-CX Switch Software Documentation Portal	https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
HPE Aruba Networking Hardware	https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm

Documentation
and Translations
Portal

HPE Aruba
Networking
software <https://networkingsupport.hpe.com/downloads>

Software
licensing and
Feature Packs <https://lms.arubanetworks.com/>

End-of-Life
information <https://www.arubanetworks.com/support-services/end-of-life/>

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help

content, include the product name, product version, help edition, and publication date located on the legal notices page.